How Much Is Per-Sample Step Placement Worth at a Fixed Budget?

Abstract

Diffusion samplers generate images by repeatedly evaluating a denoising network. Recent methods adapt the timestep schedule to each sample. We ask how much this improves over a tuned shared schedule at the same evaluation count. Across five latent diffusion models, we compare shared and per-sample schedules using an exactly solved finite-grid surrogate based on trajectory curvature. We then estimate the image-level benefit on two models by searching DDIM schedules and measuring LPIPS against 200-evaluation references generated from the same initial noise. Although different samples often favour different schedules, the mean full-rollout gain at budgets of 4 and 8 denoising evaluations corresponds to only 0.25–0.71 additional evaluations under a local-linearity assumption. In all four settings, adding one evaluation yields a larger gain. Schedule diversity alone therefore does not imply that sample-specific scheduling is worthwhile.

1 Introduction

A diffusion sampler must decide where along the reverse-time trajectory to place its denoising evaluations. Using many evaluations can improve fidelity, but also makes sampling slow. At a fixed evaluation count, the selected times form the *timestep schedule*. It is distinct from the training noise schedule and can be changed without retraining.

Most scheduling methods optimise one global schedule for every sample, using objectives such as FID [8], KID [15], a KL-divergence upper bound [10], or fidelity to a teacher solver [12]. Recent work also studies sample-specific schedules [20, 21], using a predictor to adapt evaluation times to each sample. The potential value of this approach depends on how much sample-specific schedules improve over a tuned shared schedule at the same evaluation count.

GITS [1, 2] evaluates per-sample selection using aggregate FID. INDIS [21] and Yu et al. [20] evaluate learned predictors, so their results also include prediction accuracy. We complement these studies by pairing each image's searched schedule with a tuned shared schedule.

Our contributions are:

- We prove that positive rescaling of a sample's error profile does not change its optimal schedules. At fixed budget, only temporal shape can affect placement.
- Across five 256×256 latent diffusion models, we compute exact shared and per-sample optima of the finite-grid surrogate. Although optimal meshes often differ across samples, the largest median excess surrogate cost is 3.13%.
- In full DDIM rollouts on church and bedroom, the mean benefit of per-image scheduling is smaller than the gain from one additional denoising evaluation.

Figure 1 (Appendix B.1) and Sections 2–3 present the profiles and surrogate analysis; Section 4 gives the full-rollout experiment, and Section 5 distinguishes placement from budget and stopping. Appendix C examines timestep rounding.

2 Scale Cancellation at Fixed Budget

To compare schedules without a model run for every candidate, our surrogate sums estimated local-error magnitudes over a candidate mesh. Let $x_k(t)$ denote sample k's latent trajectory. We use B for intervals on the retained surrogate domain and N for denoising evaluations in a full rollout (Section 4). Candidate meshes use the finite retained grid $\mathcal{G}$ constructed in Section 3, with lengths measured in nominal unrounded-grid coordinates (Appendix B.2.1); only model evaluations use rounded timesteps. Every mesh includes the first and last points, so only the $B-1$ interior nodes vary. We write $M = \{t_0, \dots, t_B\}$, with $h_i = |t_{i+1} - t_i|$. For a method of order p, we model the leading local-error magnitude on step i as $c_k(t_i)h_i^{p+1}$, where $c_k(t)$ is a nonnegative sample-specific coefficient. Summing these terms gives the surrogate

$$E_k(M) = \sum_{i=0}^{B-1} c_k(t_i) h_i^{p+1}. \tag{1}$$

The power $p+1$ follows from the standard local-error expansion for an order-p method; Appendix A gives the short derivation. Summing the magnitudes

is a modelling choice: it omits propagation and cancellation through later steps (Appendix B.4.2). We call the tabulation of c_k on $\mathcal{G}$ sample k's error profile, or simply its profile.

Profiles can differ in two ways: their overall scale and their shape over time. If profiles differ only in scale, write $c_k(t) = a_k\phi(t)$, where ϕ is their common temporal shape. A larger a_k means more local error everywhere but leaves the relative pattern over time unchanged. For an arbitrary profile c on $\mathcal{G}$, let $E_c(M)$ denote the right-hand side of Eq. (1) with c_k replaced by c, and define the profile sum as $S(c) = \sum_{t\in\mathcal{G}} c(t)$.

Proposition 1 (Scale cancellation)**.** *Let c be any nonzero, nonnegative profile on $\mathcal{G}$. At fixed B, the set of optimal meshes depends on c only through its normalised profile $c/S(c)$:*

$$\arg\min_M E_c(M) = \arg\min_M E_{c/S(c)}(M).$$

Equivalently, replacing c by $a\,c$ for any $a > 0$ does not change the set of optimal meshes.

Proof. Let $a\,c$ denote the profile obtained by multiplying every value of c by a. Linearity in the profile gives $E_{a\,c}(M) = aE_c(M)$. Because $z \mapsto az$ preserves the ordering of objective values for $a > 0$, the two objectives have the same optimal meshes. Taking $a = 1/S(c)$ gives the normalised form. □

Each profile may be normalised without changing its optimal meshes. Section 3 uses the normalised profiles to define the shared objective; Appendix A compares exact dynamic programming [1, 14] with the classical asymptotic formula.

3 How Much a Shared Mesh Costs

We evaluate five released 256×256 models of Rombach et al. [9]: unconditional LSUN-church and LSUN-bedroom [19], CelebA-HQ [6], FFHQ [7], and class-conditional ImageNet [4]. For each, we draw 256 deterministic DDIM [$\eta = 0$; 11] trajectories on a common 200-step grid; the ImageNet run contains 225 distinct class labels.

DDIM is a first-order sampler, so we set $p = 1$ in Eq. (1); its leading one-step error is then $O(h^2)$. Because $c_k(t)$ is unobserved, we approximate its temporal shape with the Euclidean norm of the second difference, assigned to its centre state:

$$\hat{c}_{k,j-1} = \|x_{k,j} - 2x_{k,j-1} + x_{k,j-2}\|.$$

We measure each profile once along its 200-step trajectory and hold it fixed when evaluating candidate meshes under the surrogate. We do not run the model for candidate meshes. Each trajectory has 201 latent states, yielding 199 three-point differences centred at rounded timesteps 994 through 5. For equally spaced states, the second difference approximates the second derivative up to a positive grid-spacing factor. Because this factor only rescales the profile, Proposition 1 ensures that the optimal meshes cannot change.

We remove two parts of the grid that do not measure curvature. First, the final DDIM step ends at $t = 0$ and produces a large second difference in every sample. We observe that the final nine values form a common terminal spike and, averaged within each model, account for 36–63% of the sum of all 199 second-difference magnitudes. We exclude this shared endpoint feature from the main surrogate analysis so that it does not dominate the normalised profiles. Appendix B.4.3 varies the trim from 0 to 30 points; the main cost pattern remains stable.

Second, rounding the integer grid leaves one size-4 step among 199 of size 5. As shown in Appendix C, unequal adjacent steps cause the second difference to include a first-derivative term. We drop the two affected points; they are adjacent and form one gap. If retained as candidates, neither appears in any optimum in the ten model–budget combinations, so Table 1 is unchanged (Appendix C). That leaves $199 - 9 - 2 = 188$ retained points spanning rounded timesteps 994 to 50, excluding the final stretch to $t = 0$. Here, B counts intervals on this surrogate domain. Removed entries are profile measurements, not denoising evaluations. The retained points form $\mathcal{G}$. We normalise each measured profile pointwise as

$$\tilde{c}_{k,j} = \frac{\hat{c}_{k,j}}{\sum_{\ell \in \mathcal{G}} \hat{c}_{k,\ell}}, \qquad j \in \mathcal{G}.$$

We call the denominator $S(\hat{c}_k)$ the profile sum, or overall magnitude. By Proposition 1, this normalisation does not change its optimal meshes.

For each model, let $K = 256$ be the number of measured samples and let $\bar{c} = K^{-1} \sum_{k=1}^{K} \tilde{c}_k$ be their mean normalised profile. The average cost of a shared mesh can be computed either across all K sample profiles or once using their mean profile:

$$\frac{1}{K} \sum_{k=1}^{K} E_{\tilde{c}_k}(M) = E_{\bar{c}}(M),$$

where the equality follows from the linearity of $E_c(M)$ in c. The profile $\bar{c}$ is an aggregate of all K samples, not an individual sample. The shared and

sample-specific meshes are

$$M^\star_{\mathrm{sh}} \in \arg\min_M E_{\bar{c}}(M), \qquad M^\star_k \in \arg\min_M E_{\tilde{c}_k}(M).$$

We find both by dynamic programming. Table 1 summarises the 256 per-sample excess values

$$g_k = 100\left(\frac{E_{\tilde{c}_k}(M^\star_{\mathrm{sh}})}{E_{\tilde{c}_k}(M^\star_k)} - 1\right).$$

Using unnormalised profiles gives more weight to samples with larger profile sums, but selects the same shared mesh in all ten settings (Appendix B.2.1). Table 1 has small median costs but heavy tails, plotted in Figure 2. Church $B = 8$ has 165 unique optima with 3.13% median excess, while ImageNet $B = 4$ has 206 samples on the shared mesh but a 94% maximum. Mesh diversity need not imply a large median gain, nor does low diversity rule out a costly tail.

Table 1: Excess normalised-surrogate cost (%) of the exact shared optimum relative to exact per-sample optima on the retained 188-point grid (timesteps 994–50). Rows summarise 256 samples; B counts intervals. Final columns count distinct optima and samples matching the shared optimum.

model	B	median	p90	p99	max	unique meshes	shared
LSUN-church	4	2.11%	8.12%	26.22%	28.30%	66	15
LSUN-church	8	3.13%	8.74%	19.12%	24.45%	165	10
LSUN-bedroom	4	0.21%	7.83%	36.73%	72.35%	14	38
LSUN-bedroom	8	0.32%	5.96%	30.50%	60.85%	47	14
CelebA-HQ	4	0.44%	2.77%	11.43%	21.09%	32	20
CelebA-HQ	8	1.30%	5.82%	11.87%	19.33%	118	19
FFHQ	4	0.02%	0.25%	0.70%	13.77%	12	105
FFHQ	8	0.01%	0.37%	1.58%	14.11%	33	122
ImageNet	4	0.00%	0.45%	18.57%	94.47%	17	206
ImageNet	8	0.34%	1.63%	15.41%	83.52%	37	41

4 Estimating Placement Benefit

In Table 1, we exactly optimise B-interval meshes under the surrogate using 200-step profiles, not image quality. At rollout budget N, we run each candidate DDIM ($\eta = 0$) schedule and compare its final image by LPIPS [23] with a same-noise, 200-evaluation reference. Each schedule uses the same ($K = 256$) initial-noise samples per model. For a schedule M and initial noise $z_T^{(k)}$, define

$$Y_k(M) = \mathrm{decode}\Big(\mathrm{DDIM}_M(z_T^{(k)})\Big), \quad y_k^{\mathrm{ref}} = Y_k(M^{\mathrm{unif}}_{200}), \quad D_k(M) = \mathrm{LPIPS}\big(Y_k(M), y_k^{\mathrm{ref}}\big).$$

The uniform M^{unif}_{200} defines y_k^{ref} as a model output from the same initial noise $z_T^{(k)}$, not ground truth. Let $\bar{D}(M)$ average over k. Starting from uniform,

cyclic coordinate descent [17] returns $\widehat{M}_{\mathrm{sh},N}$ for $\bar{D}$ and then, initialised there, $\widehat{M}_{k,N}$ for D_k. Table 2 reports

$$\Delta_N = \frac{1}{K}\sum_{k=1}^{K}\left[D_k(\widehat{M}_{\mathrm{sh},N}) - D_k(\widehat{M}_{k,N})\right], \qquad R_N = \frac{\Delta_N}{\bar{D}(\widehat{M}_{\mathrm{sh},N}) - \bar{D}(\widehat{M}_{\mathrm{sh},N+1})}.$$

R_N divides the per-image LPIPS gain by the gain from increasing the shared budget from N to $N{+}1$, with the $N{+}1$ schedule searched separately. Table 2 calls this "recovered NFE"; under local linearity, $R_N < 1$ means placement helps less than one additional evaluation. A hat denotes a coordinate-descent output, not a certified global optimum. Following LD3 [12], the objective is fidelity to a high-budget model output, not distributional quality; Appendix B.3.1 gives the search and bootstrap protocols.

All four estimates and intervals are below one additional evaluation; the widest reaches 0.83. At church $N = 8$, 255 of 256 images improve with 255 distinct schedules. Three church $N = 8$ searches still improve when all adjustable timesteps are checked for the tenth time and are then stopped, so the 0.713 estimate may be understated. Appendix B.3.3 measures how far coordinate descent falls short of the exact surrogate optimum. The LPIPS optimum is unknown, so the full-rollout results report only the gains found by coordinate descent. They do not establish the maximum possible per-image gain or evaluate a predictor.

5 Related Work: Placement, Budget, and Stopping

Adaptive scheduling can change where a fixed number of evaluations are placed (*step placement*), assign different counts to different samples (*budget allocation*), or stop at sample-dependent times (*stopping*). We study step placement at a fixed evaluation count.

Shared schedules have been optimised for different objectives. ART minimises aggregate Euler discretisation error on a finite grid [5]; Score-Optimal derives a shared schedule in the dense-step limit from a Stein-score transport cost [16]; and denoising-growth bounds yield certified Euler schedules and finite-grid block partitions [13].

Per-sample methods ask a different question. The journal extension of GITS [1, 2] reports worse aggregate FID than the best shared schedule at 5 and 6 NFE, but slightly better FID at 8 and 10 NFE. These aggregate scores do not reveal each sample's gain over the shared schedule. INDIS predicts input-dependent timesteps, shifts and output scales [21], while Yu

et al. [20] learn prompt- and noise-conditioned schedules for a frozen sampler. Their image experiments combine schedule quality with predictor accuracy and report distributional, alignment or reward-based metrics. We instead compare searched schedules directly, using a fixed same-noise reference for each sample.

On a two-dimensional synthetic problem, INDIS also compares one globally optimised schedule with two per-instance alternatives: schedules optimised separately for each instance and schedules produced by a learned model. Both alternatives have lower endpoint error.

Other methods change the evaluation count. TPDM predicts the next noise level and stops at a threshold [18]; it matches GITS in FID at a comparable SD1.5 step count, though sampler steps are not NFE. AdaDiff selects one of ten step counts from the prompt embedding [22]. These methods address stopping or budget allocation, outside our fixed-budget analysis.

6 Conclusion and Limitations

Across five models and $B \in \{4, 8\}$, we find that the selected optimum differs from the shared mesh for a majority of samples in nine of ten settings. However, the largest median excess surrogate cost is 3.13%, while mean full-rollout LPIPS gains correspond to only 0.25–0.71 additional evaluations under local linearity. Although samples often favour different schedules, this diversity does not by itself imply a meaningful benefit from per-sample scheduling. A fair comparison should therefore use a tuned shared schedule at the same evaluation count.

All four full-rollout experiments support this conclusion, but they cover only deterministic DDIM on LSUN-church and LSUN-bedroom at two budgets. Stochastic and text-conditioned sampling remain untested. The four numerical estimates also vary with the images included: compared with runs on the first 64 images, estimates from 256 images range from 9% lower to 22% higher. Nevertheless, both sample sizes give the same qualitative result: per-image scheduling improves LPIPS less than adding one evaluation to the shared schedule.

The surrogate isolates step placement but does not model the full sampling process. It uses profiles measured on 200-step trajectories to compare meshes with $B \in \{4, 8\}$ intervals, but omits error propagation, local-error direction, schedule-induced trajectory changes, and the excluded terminal region, where every model's mean profile is largest.

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A Surrogate Checks and Optimality Geometry

A.1 Origin of the Surrogate

When an order-p method starts a step from the exact state, its local error has the expansion

$$\tau_{k,i} = A_k(t_i)h_i^{p+1} + O(h_i^{p+2}).$$

Its leading magnitude is $\|\tau_{k,i}\| \simeq c_k(t_i)h_i^{p+1}$ after defining $c_k(t_i) = \|A_k(t_i)\|$. For example, forward Euler omits the $\frac{1}{2}x_k''(t_i)h_i^2$ term in the Taylor expansion, motivating the second-difference proxy used for first-order DDIM in Section 3. Equation (1) then makes the additional approximation of summing these leading magnitudes; it is not a formula for propagated global error. Schedule-dependent terms omitted by the DDIM proxy are noted in the limitations.

A.2 Exact Dynamic Program

Let $u_0 < \cdots < u_{L-1}$ be the retained nominal-grid positions ($L = 188$), and let $V_b(s)$ be the minimum cost of reaching u_s from u_0 in exactly b intervals. Then

$$\begin{aligned} w_c(r,s) &= c(u_r)(u_s - u_r)^{p+1}, \\ V_0(0) &= 0, \qquad V_0(s>0) = \infty, \\ V_b(s) &= \min_{0\le r<s} \{V_{b-1}(r) + w_c(r,s)\}, \quad b = 1,\dots,B. \end{aligned}$$

Each transition from u_r to u_s is charged at its starting, noise-side endpoint u_r. The exact optimum is $V_B(L-1)$; predecessor pointers recover its mesh. We use $c = \bar{c}$ for the shared mesh and $c = \tilde{c}_k$ per sample, requiring $O(BL^2)$ time and $O(BL)$ memory for each profile.

A.3 Scale and Shape Checks

Multiplying one profile by values from 0.25 to 100 leaves its dynamic-programming mesh bit-identical, as Proposition 1 requires. Reordering the same values in time can move the mesh. At $B = 8$, reversing a diffusion-like synthetic profile moves one node by 37% of the interval. The classical equidistribution formula $h^*(t) \propto c(t)^{-1/(p+1)}$ [3] recommends shorter intervals where the profile is larger. For example, when $p = 1$, multiplying $c(t)$ by four halves the recommended interval length. This is a many-step approximation: it assumes enough short intervals that the profile changes little

within each one. With only $B = 4$ intervals, each interval covers a large part of the trajectory, so that local approximation can give a poor complete mesh. Directly evaluating its predicted mesh gives objective values 3% and 125% above the exact discrete optimum on our two non-constant profiles. The latter mesh therefore costs 2.25 times the optimum. Because this error is profile-dependent and can be large at low B, we use exact dynamic programming.

A.4 Optimality Geometry

Because $E_c(M)$ is linear in c, the profiles for which a given mesh M is optimal form $\{c \geq 0 : E_c(M) \leq E_c(M') \text{ for every } M'\}$, an intersection of finitely many half-spaces through the origin and hence a polyhedral cone. These cones cover the nonnegative profile space. The comparison hyperplanes $E_c(M) = E_c(M')$ refine them into polyhedral cells on whose relative interiors the set of optimal meshes is constant. Proposition 1 follows because every cone is closed under positive scaling. Proportionality to $\bar{c}$ is sufficient, but not necessary, for a sample to prefer the shared mesh. A profile with a unique optimal mesh lies away from every hyperplane that would tie it to another mesh, although the profile may still lie on the orthant boundary. This geometry applies to the samples counted in the "shared" column of Table 1.

B Profile Measurement Details

B.1 Mean Profiles

For each sample, we divide its 188 retained profile values by their sum. The result therefore describes where along the trajectory that sample's proxy is large, not its absolute error magnitude. A uniform profile would assign $100/188 = 0.53\%$ to every retained point. Values above this level indicate regions where the surrogate places relatively more weight and therefore favours shorter steps.

B.2 Measurement and Baseline Checks

B.2.1 Grid and weighting conventions

Mesh lengths use positions in the untrimmed 199-point nominal grid. Removing two adjacent candidates leaves one interval spanning three nominal-grid steps; only model evaluations use rounded timesteps. Both dynamic

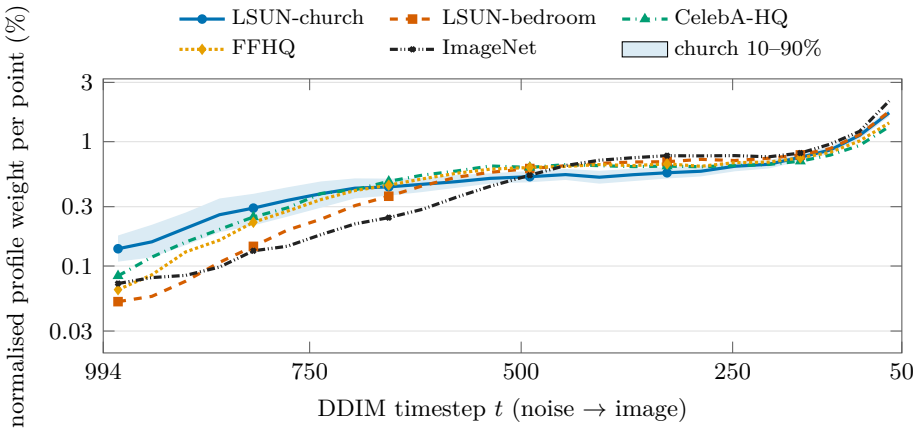


Figure 1: Mean normalised profiles over 256 samples per model. Each sample's 188 retained weights sum to 100%. For display, consecutive points are grouped into 24 bins and the mean weight per original point is plotted; the analysis uses all unbinned points. The vertical axis is logarithmic. The shaded LSUN-church band spans the 10th–90th percentiles across samples. All five mean profiles place more relative weight near the retained image-side endpoint. The band shows sample variation, not a confidence interval.

programs share the retained endpoints and use normalised profiles. Optimising the unnormalised mean profile gives more weight to samples with larger profile sums. This approach produces the same shared mesh across all ten model–budget combinations.

B.2.2 Held-out shared baseline

Table 1 fits the shared mesh and evaluates the median-excess quantity, g_k, on the same 256 profiles. To check whether this overlap makes the results look favourable, we randomly split the profiles in each cell into two groups of 128, fit the shared mesh using one group, and evaluated g_k on the other. Across 20 random splits, the held-out median g_k was, on average, within 0.04 percentage points of the corresponding Table 1 value in every cell. Even in the worst split, the median increased to only 3.8%, compared with the largest Table 1 value of 3.13%.

B.3 Full-Rollout LPIPS Search

B.3.1 Search protocol

Each row in Table 2 compares one shared schedule with 256 per-image schedules at the same budget. "Improved" counts images whose per-image schedule lowers LPIPS; "unique schedules" counts distinct returned schedules, including any unchanged shared schedule. Each search repeatedly tests candidate positions for every adjustable timestep. It stops when none improves the schedule, or after doing this ten times. Three of the 1,024 per-image searches still improve on the tenth repetition and are then stopped. After the searches, we draw 200,000 bootstrap resamples of the 256 images with replacement, using NumPy seed 0, while keeping each image's shared and per-image scores paired. For each resample, we recompute R_N while holding the searched schedules and the $N{+}1$ denominator fixed. We use the 2.5th and 97.5th percentiles and round the endpoints outward to two decimals. These conditional intervals reflect only variation in the observed image-level gains; they exclude search and denominator uncertainty and do not measure generalisation to unseen images. Algorithm 1 gives the search procedure.

Table 2: Per-image LPIPS advantage recovered by coordinate descent. Raw ΔLPIPS is mean LPIPS under the shared schedule minus the mean after per-image search; "recovered NFE" divides it by the separately searched $N{+}1$ improvement. Intervals are conditional 95% bootstrap intervals.

model	N	raw ΔLPIPS	recovered (NFE)	cond. 95% interval	improved	unique schedules
LSUN-church	4	0.0218	0.255	[0.22, 0.29]	238/256	176
LSUN-church	8	0.0199	0.713	[0.60, 0.83]	255/256	255
LSUN-bedroom	4	0.0173	0.247	[0.20, 0.30]	205/256	124
LSUN-bedroom	8	0.0196	0.691	[0.61, 0.78]	252/256	246

B.3.2 Changing the Search Limit and Number of Images

On the 64-image subset, raising the maximum from four to ten repetitions changes the recovered-NFE ratio by at most 2.1%, and by at most 0.2% in the two cells where four repetitions already reach the no-change condition. This does not establish global optimality: a stronger optimiser could affect the numerator and the separately searched $N + 1$ denominator differently, and the per-image searches are initialised from the shared-N solution, so the resulting effect on R_N is unknown. The 256-image comparison in Section 6 is not an independent replication, since the 256-image runs include the first 64 images.

Algorithm 1: Coordinate-descent searches used for Table 2.

Data: scores $\bar{D}, D_1, \ldots, D_K$ and uniform schedules U_N, U_{N+1}
Result: $\widehat{M}_{\mathrm{sh},N}$, $\widehat{M}_{\mathrm{sh},N+1}$ and $\widehat{M}_{1,N}, \ldots, \widehat{M}_{K,N}$
Function $\texttt{CD}(f, M)$**:**
  $q \leftarrow f(M)$
  for *at most ten repetitions* **do**
    $M_{\mathrm{old}} \leftarrow M$
    for *each interior coordinate j, in order* **do**
      $\mathcal{C}_j \leftarrow$ up to 12 legal positions between its neighbours
      for *each $v \in \mathcal{C}_j$, in order* **do**
        $q_v \leftarrow f(M_{j \leftarrow v})$
        if $q_v < q - 10^{-9}$ **then**
          $M_j \leftarrow v$; $q \leftarrow q_v$
    if $M = M_{\mathrm{old}}$ **then**
      break
  return M
$\widehat{M}_{\mathrm{sh},N} \leftarrow \texttt{CD}(\bar{D}, U_N)$
$\widehat{M}_{\mathrm{sh},N+1} \leftarrow \texttt{CD}(\bar{D}, U_{N+1})$
for $k = 1, \ldots, K$ **do**
  $\widehat{M}_{k,N} \leftarrow \texttt{CD}(D_k, \widehat{M}_{\mathrm{sh},N})$

B.3.3 Coordinate-descent calibration

We can measure search error on the surrogate because its optimum is known exactly. For 32 profiles per model, we run coordinate descent from the shared mesh while allowing every legal position, then compare its solution with the dynamic-programming optimum. Among the 11–32 samples per cell for which the shared mesh is suboptimal, mean suboptimality is 0.04%–1.56%, with a worst case of 6.21% for church at $B = 8$. This is a favourable benchmark: in the surrogate objective, moving one node affects only its adjacent steps, whereas in the full rollout, changing one node affects all subsequent states.

B.4 Profile Shape and Sharing Cost

B.4.1 Shape agreement and cost

By Proposition 1, only temporal shape can affect the optimal mesh. Table 3 shows that, in every model, the first percentile

Table 3: Profile magnitude and shape agreement at 188 retained points (256 samples per model). Magnitude is the within-model max/min ratio of profile sums; the 1% tail contains the least-correlated 1% of 32,640 normalised-profile pairs. Church controls rescale one shape by 1–4 or permute it in time.

	magnitude	mean ρ	1% tail	min
LSUN-church	1.70×	+0.977	+0.910	+0.820
LSUN-bedroom	1.56×	+0.975	+0.926	+0.882
CelebA-HQ	1.58×	+0.972	+0.905	+0.828
FFHQ	1.56×	+0.986	+0.951	+0.903
ImageNet	2.11×	+0.993	+0.952	+0.907
control: magnitude only	4.00×	+1.000	+1.000	+1.000
control: time-permuted	1.70×	+0.001	−0.156	−0.239

of pairwise correlations
exceeds +0.90. After
subtracting each profile's
average over time, the
mean temporal pattern
accounts for 96.4–99.0%
of the profiles' squared
signal. Most of the measured shape is common across samples.
But similar profiles can still favour different meshes. What matters for us is the extra surrogate cost of requiring every sample to use one shared mesh. Correlation compares the whole profile, whereas the cost depends on the nodes selected by the mesh and the lengths of the intervals between them. Two artificial examples show this distinction. First, shifting each profile left or right by at most one grid point raises the 1% correlation tail above that of the real church profiles (+0.967 versus +0.910), but also raises median sharing cost (3.4% versus 3.1%). Second, changing how much error lies in the first versus last third of the trajectory lowers the correlation tail to +0.773, yet median sharing cost is only 1.8%. Higher correlation does not reliably imply lower sharing cost. Table 3 describes similarity; Table 1 directly measures the cost of sharing.

B.4.2 Directional Alignment of Local Errors

Equation (1) adds the magnitudes of local-error vectors, which is most plausible when those vectors tend to point in similar directions. Across the five models, the mean cosine between consecutive error vectors is +0.836–+0.929; at a separation of 50 measurement points it remains +0.40–+0.61. Fifty points is longer than the median interval of the shared mesh in every model–budget cell, so the positive alignment supports summing magnitudes rather than treating the errors as orthogonal. Replacing the ordinary sum by root-sum-of-squares leaves the largest median sharing cost near 3.1% (3.13% versus 3.02%), but raises the largest p90 from 8.74% to 13.95%. The typical cost is stable, while the upper tail depends more strongly on the aggregation rule.

B.4.3 Changing the Endpoint Trim

Section 3 removes the final nine profile measurements because they form a large spike shared by all samples. To test whether the result depends on choosing exactly nine, we repeat the analysis after removing 0, 3, 5, 9, 14,

20 or 30 measurements. The correlation threshold for the least-similar 1% of profile pairs changes substantially, falling from +0.995 to +0.693. The cost result changes less: across these settings, the largest cell median is 1.9%–4.0% and the largest p90 is 8.0%–9.9%. The small typical sharing cost does not depend strongly on removing exactly nine measurements, although the correlation value does.

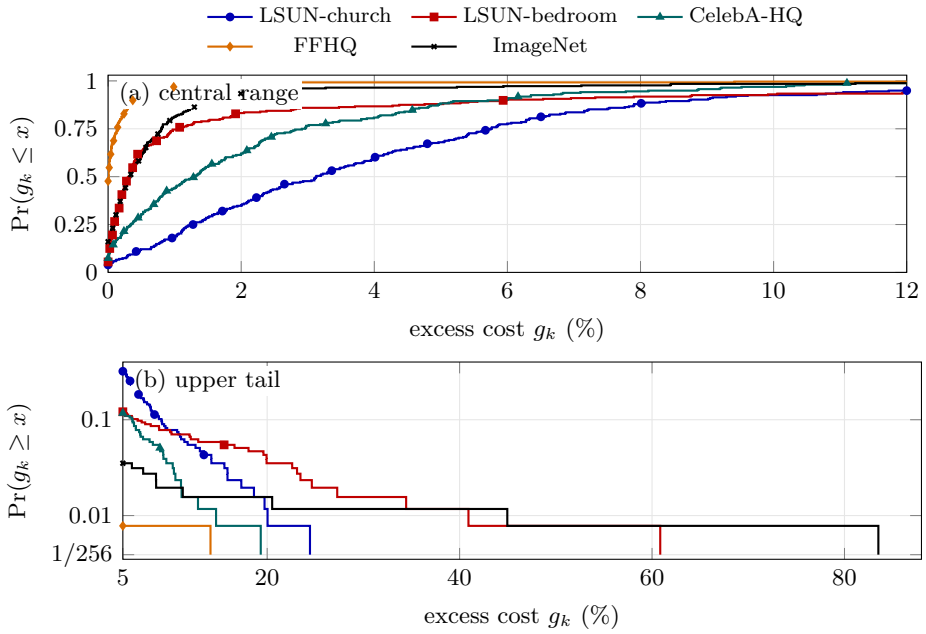


Figure 2: Distribution of excess shared-mesh cost g_k at $B = 8$ over 256 samples per model. Panel (a) shows the empirical cumulative distribution over the central 0–12% range. Panel (b) shows the fraction at or above each cost from 5% through the observed maxima, with a logarithmic vertical axis. Most samples cluster near zero, while a few form long upper tails.

B.5 Measurement-Grid Checks

B.5.1 Grid refinement

Repeating the first-order measurement at 50, 100 and 200 steps on the same 64 images gives 1% tails of +0.909, +0.910 and +0.895 on church, and +0.946, +0.939 and +0.929 on bedroom, with the trim held at 5% of each grid. These use 64 images, so the 200-step church entry differs from

the 256-sample value in Table 3. All three values stay within 0.02 of each other for both models; bedroom decreases monotonically while church does not, so this shows high agreement rather than clear grid convergence. The finite-difference scale also behaves as expected: from 50 to 100 to 200 steps, the median over all samples and the middle fifth of church profile points falls by factors of 0.256 and 0.255, matching the h^2 scaling of a second difference.

C A Rounding Artifact in the Measurement Grid

We form $n + 1$ equally spaced points from 999 to 0 and round them using NumPy's round-to-nearest-even convention. The resulting grid is not uniform: at $n = 200$ it holds one size-4 step among 199 of size 5, at the midpoint. The $n = 50$ and $n = 100$ grids each have one short step as well. Writing h_j for the step ending at grid point j, the second difference expands as

$$x_j - 2x_{j-1} + x_{j-2} = (h_j - h_{j-1})\, x' + \tfrac{1}{2}\big(h_j^2 + h_{j-1}^2\big)x'' + O(h^3).$$

When adjacent steps match, $h_j - h_{j-1} = 0$, the first-derivative term vanishes, and the operator measures curvature as intended. At the short step, the first-derivative term remains. At the two affected centres, timesteps 504 and 500, the adjacent step sizes are 4 and 5. The first-derivative coefficient therefore has magnitude 1, while the curvature coefficient is $(4^2 + 5^2)/2 = 20.5$; the former dominates when $\|x'\| \gtrsim 20.5\|x''\|$. For every sample and model, each affected point is at least 3.5 times the mean of four nearby unaffected points, and they carry 2.4–7.0% of the summed second-difference magnitude on the 200-step grid. On the 100-step grid the share falls to 1.6–3.8%. The usual step is twice as long, so the genuine curvature term is four times larger. A third difference uses one more step and is affected more strongly; its three affected points carry 6.3–18.0% of its summed magnitude on the 100-step grid.

The artifact affects the two analyses differently. The cost table is unchanged: recomputing Table 1 with the affected points included reproduces every printed value and unique-mesh count, because no optimum in the ten cells selects either point. The correlations do change, in either direction and by as much as 0.11. On the uncorrected grid, the 1% tails are +0.923, +0.916, +0.952, +0.958 and +0.846 in the order of Table 3. Table 2 uses no finite differences and is unaffected.